MT8002 Elements of Set Theory
Chapter 4. The Zermelo–Fraenkel Axioms
Section 4.5. Axioms 7 to 9

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Preliminaries

Textbook

Derek Goldrei (1996). Classic Set Theory. A Guided Independent Study.

Convention

The numbers and page numbers assigned to chapters, examples, exercises, figures, quotes, sections and theorems on these slides correspond to the numbers assigned in the textbook.

Outline

Introduction
Inductive Sets
The Axiom of Infinite
The Construction of the Set of the Natural Numbers
The Axiom of Replacement
The Axiom of Foundation
References

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The Axiom of Replacement
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Introduction

Formal language

Recall that in ZFC the concepts of pure set and membership relation are primitive (undefined). Also recall that the universe of discourse of the theory is the pure sets.

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Classification

- Axioms stating the existence of sets
- Axioms determining properties of sets
- Axioms for building sets from other sets

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The Axiom of Infinite
The Construction of the Set of the Natural Numbers
The Axiom of Replacement
The Axiom of Foundation
References

Inductive Sets

Definition (p. 38)

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Example

$$\begin{aligned} \emptyset^+ &\doteq \emptyset \cup \{\emptyset\} \\ &\doteq \{\emptyset\}, \end{aligned}$$

$$\begin{aligned} \emptyset^{++} &\doteq (\emptyset^+)^+ \\ &\doteq \{\emptyset\}^+ \\ &\doteq \{\emptyset\} \cup \{\{\emptyset\}\} \\ &\doteq \{\emptyset, \{\emptyset\}\}, \end{aligned}$$

$$\begin{aligned} \emptyset^{+++} &\doteq (\emptyset^{++})^+ \\ &\doteq \{\emptyset, \{\emptyset\}\}^+ \\ &\doteq \{\emptyset, \{\emptyset\}\} \cup \{\{\emptyset, \{\emptyset\}\}\} \\ &\doteq \{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}\}. \end{aligned}$$

Inductive Sets

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A set y is **inductive** if

(a) $\emptyset \in y$,

(b) if $x \in y$ then $x^+ \in y$.

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Remark

Note that an inductive set is an infinite set.

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The Axiom of Infinite

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Informal description

"*There is an inductive set.*" *(p. 92)*

Category

Axiom stating the existence of set.

The Axiom of Infinite

Example

Let x be an inductive set by the Axiom of Infinite. Then $x^+, x^{++}, x^{+++}, \ldots$ are also inductive sets.

Outline

Introduction
Inductive Sets
The Axiom of Infinite
The Construction of the Set of the Natural Numbers
The Axiom of Replacement
The Axiom of Foundation
References

The Construction of the Set of the Natural Numbers

Defining the natural numbers

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Definition (p. 38)

The numeral zero is defined by $\mathbf{0} := \emptyset$.

Definition

A **numeral** $\mathbf{n}$ is the representation of a natural number n in a formal system.

Example

Some numerals from zero and successor.

$$\begin{aligned}
\mathbf{1} &:= \mathbf{0}^+ \doteq \mathbf{0} \cup \{\mathbf{0}\} && \doteq \{\mathbf{0}\},\\
\mathbf{2} &:= \mathbf{1}^+ \doteq \mathbf{1} \cup \{\mathbf{1}\} \doteq \{\mathbf{0}\} \cup \{\mathbf{1}\} && \doteq \{\mathbf{0},\mathbf{1}\},\\
\mathbf{3} &:= \mathbf{2}^+ \doteq \mathbf{2} \cup \{\mathbf{2}\} \doteq \{\mathbf{0},\mathbf{1}\} \cup \{\mathbf{2}\} && \doteq \{\mathbf{0},\mathbf{1},\mathbf{2}\}.
\end{aligned}$$

Example

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\end{aligned}$$

The same numerals from the empty set and successor.

$$\begin{aligned}
\mathbf{1} &\doteq \emptyset^+ && \doteq \{\emptyset\},\\
\mathbf{2} &\doteq \emptyset^{++} && \doteq \{\emptyset, \{\emptyset\}\},\\
\mathbf{3} &\doteq \emptyset^{+++} && \doteq \{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}\}.
\end{aligned}$$

The Construction of the Set of the Natural Numbers

Definition (p. 40)

Let y be any inductive set. The **set of the natural numbers**, denoted $\mathbb{N}$, is the intersection of all inductive subsets of y, that is,

$$\mathbb{N} := \bigcap \{z : z \text{ is an inductive subset of } y\}, \qquad \text{(naive definition)},$$

$$\mathbb{N} := \bigcap \{z \in \mathcal{P}(y) : z \text{ is inductive}\}, \qquad \text{(ZFC definition)}.$$

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Definition (p. 40)

A **natural number** is a member of $\mathbb{N}$.

Outline

Introduction
Inductive Sets
The Axiom of Infinite
The Construction of the Set of the Natural Numbers
The Axiom of Replacement
The Axiom of Foundation
References

The Axiom of Replacement

Axiom of Replacement

$\forall x \exists y \forall y' \, (y' \in y \leftrightarrow \exists x' \, (x' \in x \wedge \phi(x', y')))$,

where $\phi(s,t)$ is a formula (possibly referring to named sets (possibly with parameters)) such that $\forall s \exists t \, (\phi(s,t) \wedge \forall t' \, (\phi(s,t') \rightarrow t' \doteq t))$.

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Informal description

"*If $\phi(s,t)$ is a class function, then when its domain is restricted to [a set] x, the resulting images form a set y.*" *(p. 92)*

Category

Axiom for building a set from other set.

The Axiom of Replacement

Example

In this example we show that from a formula (class function) $\phi(s,t)$ we can define a proper class (Exercise 4.39).

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For example, let $\phi(s,t)\colon\ s \doteq t$ be a formula (class function). From $\phi(s,t)$ we can define the proper class

$$\mathcal{A} := \{\langle s,s\rangle \,:\, s \text{ is a set and } \phi(s,s) \text{ holds}\}.$$

The Axiom of Replacement

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For example, let $\phi(s,t)\colon\ s \doteq t$ be a formula (class function). From $\phi(s,t)$ we can define the proper class

$$\mathcal{A} := \{\langle s,s\rangle \ :\ s \text{ is a set and } \phi(s,s) \text{ holds}\}.$$

Note that $\mathcal{A}$ is a proper class because the universe of sets $\mathcal{V}$ is the proper class of first coordinates of the ordered pairs it contains.

The Axiom of Replacement

Exercise 4.23

Let x be a set. Show that $\{\{y\} : y \in x\}$ is a set.

The Axiom of Replacement

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Solution

Let $\phi(s,t) : t \doteq \{s\}$ be a formula (class function).

By the Axiom of Replacement when the formula (class function) $\phi(s,t)$ is restricted to the set x there is a set formed with the images of the class function, that is, $\{\{y\} : y \in x\}$ is a set.

The Axiom of Replacement

Exercise

In this exercise we show that we can define an infinite set using the Axiom of Replacement.

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The first levels of the cumulative hierarchy of sets are defined by

$$V_{\mathbf{0}} := \emptyset,$$
$$V_{\mathbf{n}^+} := \mathcal{P}(V_{\mathbf{n}}), \text{ for } \mathbf{n} \in \mathbb{N}.$$

Show that $\{V_{\mathbf{n}} : \mathbf{n} \in \mathbb{N}\}$ is a set.

(continued on next slide)

The Axiom of Replacement

Solution

We define a formula (class function) by

$$\phi(n,x):\ \exists f\,(f \text{ is a function} \wedge \mathrm{Dom}(f) \doteq n^+ \wedge f(\mathbf{0}) \doteq \emptyset \wedge \forall k\,(k \in n \rightarrow f(k^+) \doteq \mathcal{P}(f(k))) \wedge f(n) \doteq x).$$

By the Axiom of Replacement when restricting $\phi(n,x)$ to the set $\mathbb{N}$ we get the set

$$A := \{x : \exists \mathbf{n}\,(\mathbf{n} \in \mathbb{N} \wedge \phi(\mathbf{n},x)\},$$

which is the required set, that is,

$$\{V_{\mathbf{n}} : \mathbf{n} \in \mathbb{N}\} \doteq A.$$

The Axiom of Replacement

Technical remark

Since there is an instance of the Axiom of Replacement for each formula (class function) $\phi(s,t)$ such that $\forall s \exists t\,(\phi(s,t) \wedge \forall t'\,(\phi(s,t') \rightarrow t' \doteq t))$, some authors define it as an axiomatic schema.[†]

[†] See, e.g., (Enderton 1977; Hrbacek and Jech [1978] 1999; Button 2021).

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Category

Axiom determining a property of sets.

The Axiom of Foundation

Axiom of Foundation

$\forall x\,(x \neq \emptyset \rightarrow \exists y\,(y \in x \wedge x \cap y \doteq \emptyset))$.[†]

Category

Axiom determining a property of sets.

[†]See the corrections to the textbook in the course homepage.

The Axiom of Foundation

Exercise 4.41(a)

Let z by a set. Show that $z \notin z$.

The Axiom of Foundation

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Let z by a set. Show that $z \notin z$.

Solution

The singleton $\{z\}$ is a set by the Axiom of Pairs whose unique element is z. By the Axiom of Foundation

$$\{z\} \cap z \doteq \emptyset.$$

Therefore, $z \notin z$ (otherwise $\{z\} \cap z \neq \emptyset$).

The Axiom of Foundation

Definition

An **infinite descending $\in$-chain** is a sequence $\langle x_n \rangle$ such that $x_{n+1} \in x_n$, for all $n \in \mathbb{N}$, that is,

$$\cdots x_{n+1} \in x_n \in \cdots \in x_2 \in x_1 \in x_0.$$

The Axiom of Foundation

Theorem 4.3 (No infinite descending $\in$-chains)

Let $\langle x_n \rangle$ be a sequence. The sequence cannot form an infinite descending $\in$-chain.

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Let $\langle x_n \rangle$ be a sequence. The sequence cannot form an infinite descending $\in$-chain.

Proof (by contradiction)

Suppose that $\langle x_n \rangle$ forms an infinite descending $\in$-chain.

Let $x := \{x_n : n \in \mathbb{N}\}$. By the Axiom of Foundation exists y such that $x \cap y \doteq \emptyset$.

Since $y \in x$, then $y \doteq x_n$ for some n. Since $x_{n+1} \in x$ and $x_{n+1} \in y$ then $x \cap y \not\doteq \emptyset$.

Contradiction!

Outline

Introduction
Inductive Sets
The Axiom of Infinite
The Construction of the Set of the Natural Numbers
The Axiom of Replacement
The Axiom of Foundation
References

References

Tim Button (2021). Set Theory. An Open Introduction. Fall 2021. Revision 2138a50 (2026-07-11). Independently published. URL: https://builds.openlogicproject.org/courses/set-theory/ (cit. on p. 36).

Herbert B. Enderton (1977). Elements of Set Theory. Academic Press (cit. on p. 36).

Derek Goldrei (1996). Classic Set Theory. A Guided Independent Study. Chapman & Hall (cit. on p. 2).

Karel Hrbacek and Thomas Jech [1978] (1999). Introduction to Set Theory. Third Edition, Revised and Expanded. Marcel Dekker (cit. on p. 36).